Whole-genome analysis of avian paramyxovirus 4 reveals two distinct Eurasian lineages

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Background / Aims

- **Avian paramyxovirus 4 (APMV-4)** is found sporadically in wild birds worldwide, particularly migratory waterfowl such as ducks and geese, which act as natural reservoirs.
- In our laboratory's ongoing viral surveillance of migratory waterfowl, APMV-4 was the second most frequently isolated virus after APMV-1 (which includes Newcastle disease virus).
- Surveillance studies report a broad host range for APMV-4[1,2], yet genome-level characterization remains scarce.
- We therefore performed whole-genome analysis to better understand the **genetic diversity and circulation dynamics** of APMV-4 in wild birds.

Materials / Methods

- **HA-positive, AIV M gene–negative samples** from South Korea and Russia (n=41) were confirmed by pan-paramyxovirus PCR.
- Viral RNA was amplified by **SISPA**[3] and subjected to whole-genome sequencing.
- ML phylogenies were inferred with **IQ-TREE 3**, assessing lineages by bootstrap support and sequence identity.
- Sequences were analyzed in **BEAST X** (GTR+G4, UCLN clock, Bayesian SkyGrid; two runs of 150M generations).
- Host/location phylogeography was evaluated via Bayes factors, Markov jump counts, and Markov reward times.
- Selection was analyzed in **HyPhy** (SLAC, FEL, MEME, BUSTED).

Time-scaled phylogeny

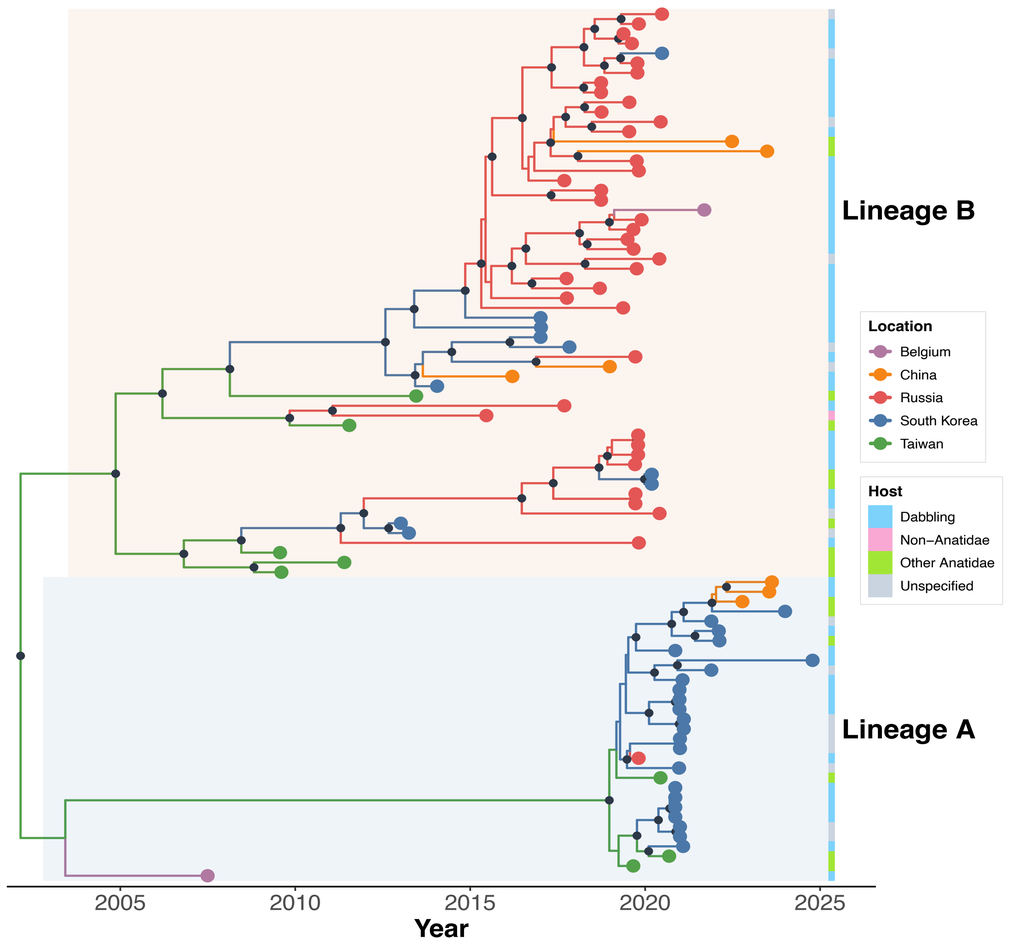


Fig 1. Time-scaled HIPSTR tree of APMV-4 genomes. Branch and tip colors indicate inferred and sampled locations. Shaded clades represent the two major Eurasian lineages, while the color strip shows host categories. Black nodes indicate pp >0.95.

Result

Lineage assignment

- APMV-4 formed two strongly supported lineages (UFBoot = 99.7%) with <90% inter-lineage identity, consistent with separate genotypes, diverging around 2002 in the time-scaled tree.

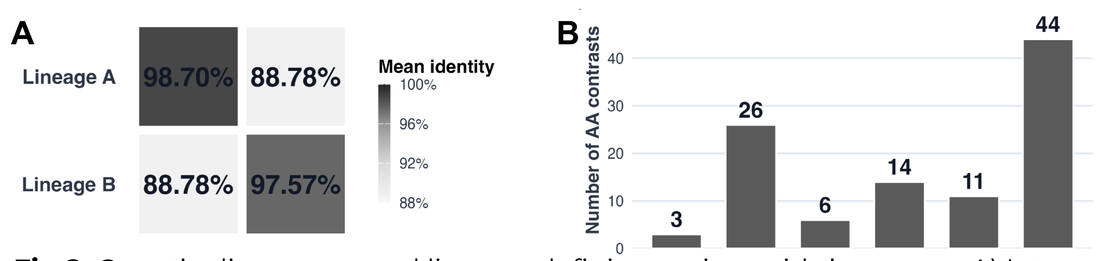


Fig 2. Genetic divergence and lineage-defining amino acid signatures. A) Intra- and inter-lineage nucleotide identity, B) Amino acid-level lineage-defining mutations

Discrete trait analysis

- Lineage A showed broader spatial diffusion across Taiwan, South Korea, Russia, and China, with South Korea and Taiwan acting as major sources and South Korea/Russia as frequent sinks.
- In contrast, Lineage B showed stronger Russia-centered persistence, with Russia and South Korea forming the main source–sink axis.



Fig 3. Lineage-specific spatial diffusion of APMV-4. Markov jump heatmaps, Markov reward time summaries, and BSSVS-supported transition maps show spatial diffusion dynamics in Lineage A (A–C) and Lineage B (D–F).

Selection analysis

- P showed greater site-wise divergence, whereas HN carried the strongest gene-wide episodic selection signal.
- 2-lineage BUSTED-PH did not support lineage-specific HN selection.

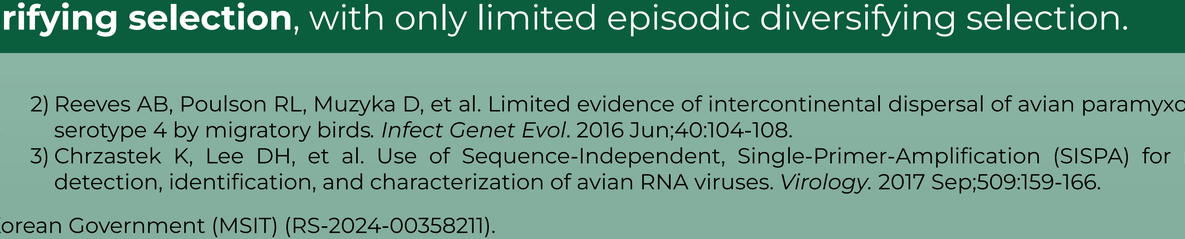


Fig 4. Selection landscape of APMV-4 coding sequences. (A) Global dN/dS values, (B) Gene-wide BUSTED analysis, (C) Site-wise SLAC/FEL/MEME scans

Limitations / Further studies

- Reference sequences were limited, as most published APMV-4 data cover only the F gene rather than the whole genome.
- Expanded sequence sampling is required to improve the reliability and resolution of APMV-4 evolutionary and phylogeographic analyses.

Conclusion

- APMV-4 comprises two major Eurasian genotypes: a **South Korea-associated Lineage A** and a **Russia-associated Lineage B**.
- Lineage A showed **East Asian circulation** with South Korea as a major source, whereas Lineage B showed **Russia-centered persistence** and **Russia–South Korea circulation**; both lineages were mainly associated with dabbling ducks and other anatidae.
- Despite lineage divergence, APMV-4 evolution is dominated by **purifying selection**, with only limited episodic diversifying selection.

References

1) Yin R, Zhang P, Liu X, et al. Dispersal and Transmission of Avian Paramyxovirus Serotype 4 among Wild Birds and Domestic Poultry. *Front Cell Infect Microbiol*. 2017 May 26;7:212.

2) Reeves AB, Poulson RL, Muzyka D, et al. Limited evidence of intercontinental dispersal of avian paramyxovirus serotype 4 by migratory birds. *Infect Genet Evol*. 2016 Jun;40:104-108.

3) Chrzastek K, Lee DH, et al. Use of Sequence-Independent, Single-Primer-Amplification (SISPA) for rapid detection, identification, and characterization of avian RNA viruses. *Virology*. 2017 Sep;509:159-166.

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